Equilibrium binding energy at fixed baryon and ADM mass in a local massive-scalar neutron-star model

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8 November 2026 · YBE-2026-11-08-a

Working manuscript prepared for checking; not peer reviewed.

Abstract

We compare the equilibrium binding energy $E_{\rm bind} = (M_B - M_{\rm ADM})c^2$ in a local massive-scalar fluid model and its scalar-free general-relativistic control. The unchanged SLy, APR4 and H4 piecewise-polytropic equations of state (EOSs) are used with $\alpha_Y = 1$ and ranges $L = 1$ and 10 km. Baryon mass is integrated over proper spatial volume; ADM mass includes the exterior scalar energy. Equal-baryon-mass and equal-ADM-mass comparisons are explicitly separated. At the baryon mass of each EOS's GR $1.4M_\odot$ configuration, the $L = 10$ km binding-energy excess is 10.850, 11.233 and 8.443×10^{51} erg, respectively. At equal ADM mass, the contrast instead measures the baryon-mass excess. Near-limit comparisons use 99% of the smaller jointly accessible causal first-rising branch limit. All 24 matched pairs satisfy this selection. Re-execution of 114 pointwise numerical controls tests finite exterior extent and tolerances. These results are conditional equilibrium diagnostics, not evidence of dynamical stability, observational detection or energy emitted in a conversion.

1 Question and model

Binding energy can compare the energy budgets of stellar equilibria, provided the baryon and gravitational masses have compatible definitions. GR hydrostatic equilibria provide a control [1]; a scalar field outside the matter surface requires an asymptotic mass definition [3]. Here the question is not whether a new stellar population has been detected, but how two distinct mass-matching prescriptions change the interpretation of a computed energy contrast.

We adopt the following phenomenological local action, rather than deriving a coupling from an experimental source fit. Throughout the field equations, $G = c = 1$ and lengths are in km; c is restored in energy conversions. The fluid number density n is defined using $g_{\mu\nu}$, with the conserved current $\nabla_\mu(nu^\mu) = 0$, and $\rho_0 = m_B n$ is the EOS rest-mass density.

$$S = \int d^4x\sqrt{-g}\left[\frac{\mathcal{R}}{16\pi} - \frac{(\nabla\phi)^2 + \phi^2/L^2}{2\alpha_Y} - e^{-\phi}\varepsilon(n)\right]. \tag{1}$$

$$ds^2 = -e^{2\nu}dt^2 + B^{-1}dr^2 + r^2d\Omega^2, \qquad B = 1 - 2m/r. \tag{2}$$

$$\tilde{\varepsilon} = e^{-\phi}\varepsilon, \qquad \tilde{P} = e^{-\phi}P, \tag{3}$$

$$\rho_\phi = \frac{B\phi'^2 + \phi^2/L^2}{2\alpha_Y}, \qquad p_{r\phi} = \frac{B\phi'^2 - \phi^2/L^2}{2\alpha_Y}, \tag{4}$$

$$m' = 4\pi r^2(\tilde{\varepsilon} + \rho_\phi), \qquad \nu' = \frac{m + 4\pi r^3(\tilde{P} + p_{r\phi})}{r^2 B}, \tag{5}$$

$$h' = -\nu' + \phi', \qquad h = \ln\frac{\varepsilon + P}{\rho_0 c^2}, \tag{6}$$

$$B\phi'' + \left(\frac{2B}{r} + B\nu' + \frac{B'}{2}\right)\phi' - \frac{\phi}{L^2} = -\alpha_Y\tilde{\varepsilon}. \tag{7}$$

The scalar equation is sourced by $e^{-\phi}\varepsilon$, not by a matter trace containing a $3P$ term. This is a specific assumed scalar–fluid model, not an identification with every trace-coupled scalar–tensor theory. The scalar-free GR control is solved separately with the same EOS. Applying the coupling to the bare tabulated $\varepsilon(n)$ is a modelling assumption whose physical interpretation remains open.

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2 EOS, mass definitions and numerical construction

The supplied implementation uses the Read et al. piecewise-polytropic representation [2]: SLy, APR4 (the AP4 parametrization in that reference) and H4, with an SLy crust. The core division densities are $10^{14.7}$ and 10^{15} g cm^{-3}. The fitted parameters $(\log_{10} P_1, \Gamma_1, \Gamma_2, \Gamma_3)$ are $(34.384, 3.005, 2.988, 2.851)$, $(34.269, 2.830, 3.445, 3.348)$ and $(34.669, 2.909, 2.246, 2.144)$, respectively, with P_1 in dyn cm^{-2}. Energy integration constants enforce continuity:

$$P/c^2 = k_i \rho_0^{\Gamma_i}, \qquad \varepsilon/c^2 = (1+a_i)\rho_0 + \frac{P/c^2}{\Gamma_i - 1}, \qquad P = n\frac{d\varepsilon}{dn} - \varepsilon. \tag{8}$$

$$M_B = 4\pi \int_0^R \frac{\rho_0(r) r^2}{\sqrt{1-2m(r)/r}}\, dr, \qquad M_{\mathrm{ADM}} = \lim_{r\to\infty} m(r), \qquad E_{\mathrm{bind}} = (M_B - M_{\mathrm{ADM}})c^2. \tag{9}$$

In Eq. (9), the mass integral is expressed in the geometric units of Eq. (2); the code converts cgs rest density to km^{-2} before integrating. The cold first law and $\varepsilon/(\rho_0 c^2) \to 1$ at low density fix the rest-mass convention. No arbitrary energy-density-to-baryon-density conversion is used. There is no lapse or $e^{-\phi}$ multiplier in M_B: those would change the conserved-baryon measure of this model. An absolute microscopic baryon count is not inferred beyond the EOS convention.

The matter surface is $h(R) = 0$. $m(R)$ is not substituted for M_{ADM}, since the scalar exterior contributes positive energy. An interior and an exterior domain are solved simultaneously by nonlinear collocation, with a regular central expansion at $r_0 = 10^{-5}$ km, continuous m, ν, ϕ, ϕ' at the fluid surface, and no surface shell. The outer radius is $r_{\mathrm{out}} = R + 18L$. The finite-domain conditions are

$$\phi'/\phi = -1/L - (1 + m_{\mathrm{out}}/L)/r_{\mathrm{out}}, \qquad \nu_{\mathrm{out}} = \tfrac{1}{2}\ln(1 - 2m_{\mathrm{out}}/r_{\mathrm{out}}). \tag{10}$$

They approximate the decaying massive-scalar solution and $\nu(\infty) = 0$. Reported ADM masses are the finite-domain approximants whose sensitivity to 12L/24L is tested below. The baseline collocation residual tolerance is 2×10^{-7}, with boundary tolerance at most 10^{-10} and at most 22000 nodes. GR has a separate scalar-free collocation system and an independent enthalpy-IVP control. Baryon quadratures use 48 and 96 Gauss–Legendre nodes separately in each EOS segment; the regular central volume is included.

Each of nine sequences has 89 fixed central-density points, $14.55 \le \log_{10}[\rho_{0c}/(\mathrm{g\,cm^{-3}})] \le 15.65$, spaced by 0.0125 dex. Stored peak searches, direct mass roots and derivative checks bring the archive to 1391 configurations. No equation, EOS parameter or numerical tolerance was adjusted to improve the reported contrasts. The source constants are $M_\odot = 1.988409870698051 \times 10^{30}$ kg and $c = 299792458$ m s^{-1}; $M_\odot c^2 = 1.787093668741279 \times 10^{54}$ erg.

3 One energy analysis, two matching prescriptions

Causality is imposed on the unchanged fitted EOS as $\max_r(dP/d\varepsilon) \le 1$, including both sides of EOS junctions. Numerical comparisons use a roundoff allowance of 10^{-10} in this dimensionless test, not an EOS-uncertainty relaxation. The selected interval ends at the earlier of the first ADM peak and first causal onset. An increasing mass is a branch-selection condition, not a demonstrated stable branch. Baryon mass is checked to be monotone on this interval.

We use four contrasts for each EOS/range pair: common $M_{\mathrm{ADM}} = 1.4M_\odot$; common M_B equal to the EOS-specific GR $1.4M_\odot$ state's baryon mass; and two separate near-limit targets at 99% of the smaller GR/Yukawa causal branch-limit ADM or baryon mass. All targets are obtained by re-solving equilibria and root-finding within that branch, without cross-fold interpolation or extrapolation. The exact pair-specific targets are in Tables 1–2.

$$\Delta E_{\mathrm{bind}}|_{M_B} = -c^2 \Delta M_{\mathrm{ADM}}, \qquad \Delta E_{\mathrm{bind}}|_{M_{\mathrm{ADM}}} = c^2 \Delta M_B, \tag{11}$$

where Δ means Yukawa minus GR. Thus a fixed-M_B positive contrast means a lower ADM energy; a fixed-M_{ADM} contrast measures a greater baryon capacity. These are complementary summaries of the same equilibrium-energy finding, not two independent discoveries. Nor is a GR-to-different-coupling comparison an allowed physical transition by itself.

For $L = 1$ km the canonical fixed-M_B excess is 0.424, 0.454 and 0.277×10^{51} erg for SLy/APR4/H4. For $L = 10$ km it is 10.850, 11.233 and 8.443×10^{51} erg. At common ADM $1.4M_\odot$, the corresponding $L = 10$ km baryon excess is 0.007612, 0.007954 and $0.005670 M_\odot$. The baryon-mass reference is EOS dependent and is not a universal $1.5M_\odot$ target.

At the matched near-limit baryon masses, the $L = 10$ km energy contrasts are 21.516, 20.839 and 18.137×10^{51} erg. Equal-ADM near-limit contrasts are different pairs, with energy differences 32.543, 31.198 and 24.514×10^{51} erg. They should not be pooled as repeated measurements of one universal conversion energy.

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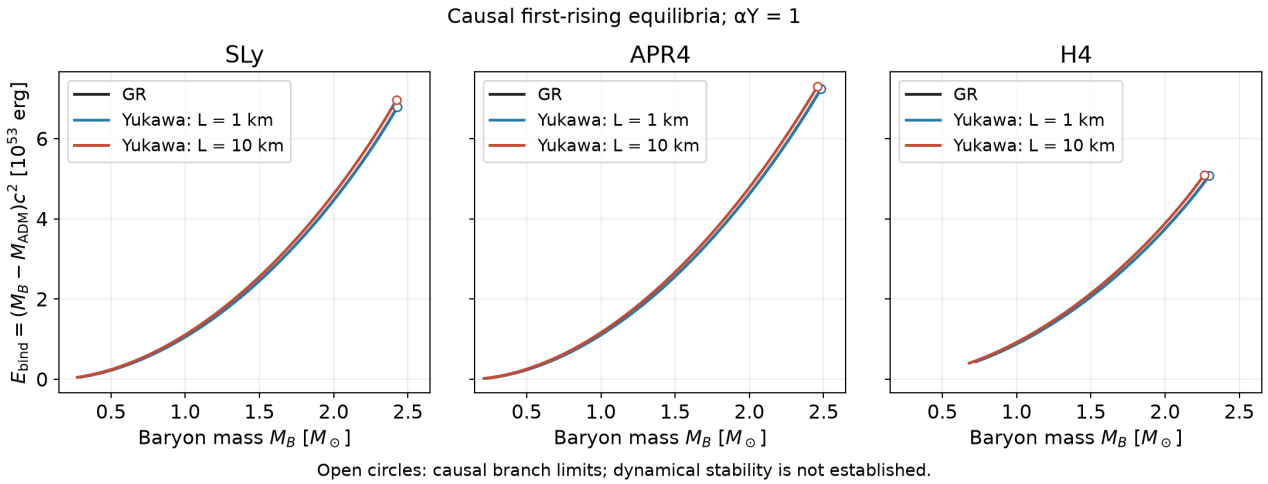


Figure 1: Binding energy versus baryon mass. Only the causal first-rising portions and their limiting points are plotted. Curves connect computed grid points; open circles denote branch limits, not stability labels. H4 begins at a larger baryon mass because the same central-density grid is used.

EOS	L	$M_{B,0}$	$E_{\mathrm{GR},0}$	$\Delta E\|_{\mathrm{ADM}}$	$\Delta E\|_B$	$\Delta M_B\|_{\mathrm{ADM}}$
SLy	1	1.546235	261.336	0.528	0.424	0.295
SLy	10	1.546235	261.336	13.604	10.850	7.612
APR4	1	1.554236	275.634	0.571	0.454	0.319
APR4	10	1.554236	275.634	14.215	11.233	7.954
H4	1	1.517822	210.558	0.331	0.277	0.185
H4	10	1.517822	210.558	10.132	8.443	5.670

Table 1: Canonical contrasts. Energies are in 10^{51} erg and L in km. $M_{B,0}$ is the baryon mass, in solar masses, of the EOS-specific GR state with $M_{\mathrm{ADM}} = 1.4M_\odot$; $E_{\mathrm{GR},0}$ is the GR reference energy. The two ΔE columns refer to different configuration pairs; all differences are Yukawa minus GR. The last column is the fixed-ADM baryon excess in $10^{-3}M_\odot$.

EOS	L	M^*_{ADM}	$\Delta E\|_{\mathrm{ADM}}$	M^*_B	$\Delta E\|_B$
SLy	1	2.026371	1.361	2.402076	0.898
SLy	10	2.012033	32.543	2.397791	21.516
APR4	1	2.055467	1.256	2.456551	0.832
APR4	10	2.029123	31.198	2.433464	20.839
H4	1	1.991272	0.919	2.272400	0.673
H4	10	1.958312	24.514	2.240432	18.137

Table 2: Near-limit targets and contrasts. Target masses are in solar masses, energies in 10^{51} erg, and L in km. Each ADM or MB target is separately 99% of the corresponding smaller, jointly reachable causal branch-limit mass. Different individual endpoints are not subtracted as if they were matched states.

EOS	L	$\log_{10}\rho_{0c}$	$M_{\mathrm{ADM,lim}}$	$M_{B,\mathrm{lim}}$	Limit
SLy	0	15.302445	2.048411	2.427987	ADM peak
SLy	1	15.302561	2.046839	2.426339	causality
SLy	10	15.302561	2.032357	2.422011	causality
APR4	0	15.160108	2.078144	2.483576	causality
APR4	1	15.160108	2.076229	2.481365	causality
APR4	10	15.160108	2.049619	2.458044	causality
H4	0	15.209106	2.013851	2.298262	ADM peak
H4	1	15.209741	2.011386	2.295354	ADM peak
H4	10	15.227781	1.978093	2.263063	ADM peak

Table 3: Causal first-rising branch limits, not established stable maximum masses. $L = 0$ denotes GR; L is in km and masses are in solar masses. Density is $\log_{10}[\rho_{0c}/(\mathrm{g\,cm^{-3}})]$. The earlier of the unconstrained ADM peak and first causal onset sets the limit.

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All H4 ADM peaks are causal ($c_s^2/c^2 = 0.515838, 0.516312, 0.529803$ for GR, 1 km and 10 km). Both SLy Yukawa peaks and all APR4 peaks are acausal and are retained only as mathematical continuations in the CSV. The SLy GR peak remains just causal (0.999774). The greatest c_s^2/c^2 among actual pair members is 0.976933.

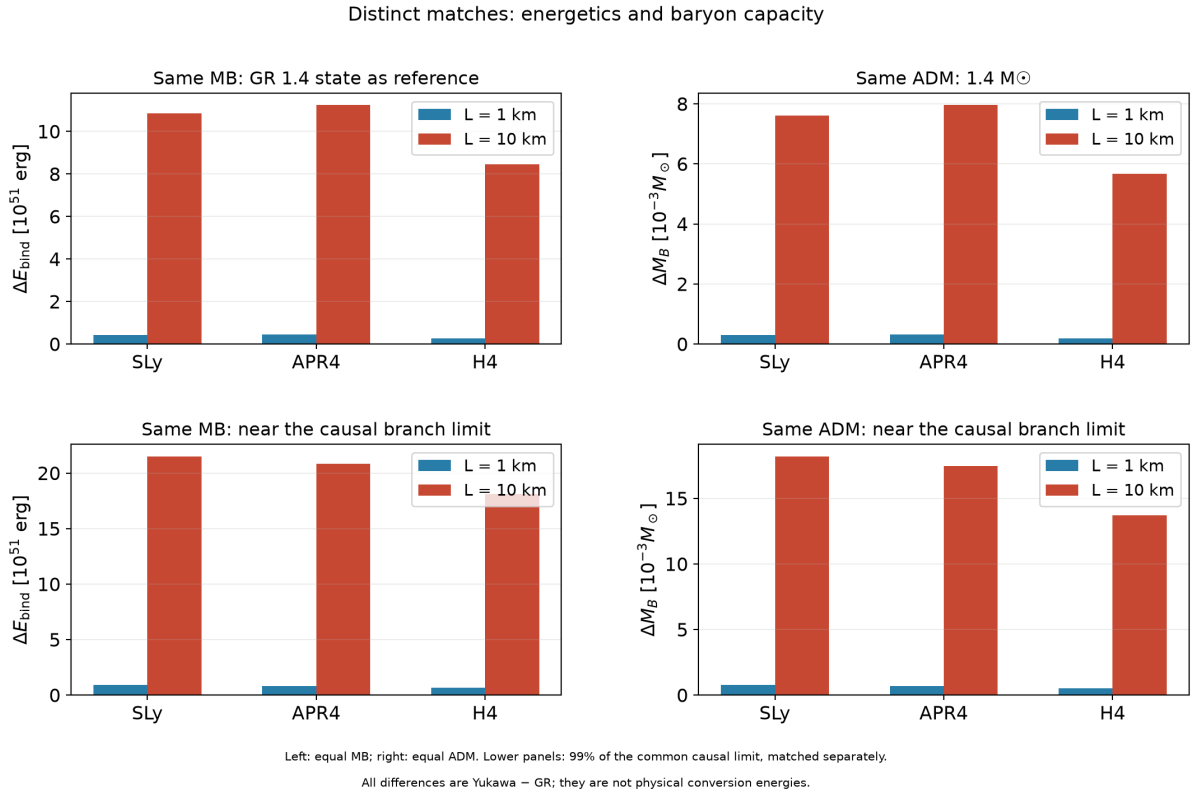


Figure 2: Mass matching changes the question. Left: energy at common baryon mass. Right: baryon capacity at common ADM mass, with a different vertical unit. The lower panels use separately defined 99% common causal-limit targets; the two columns do not select the same states.

4 Numerical checks and their scope

All supplied input hashes, mass-budget identities and energy identities were checked. The 1391 sound-speed flags were recomputed from the EOS, and all 24 target pairs were verified to remain below the applicable causal branch limit. The maximum direct mass-target residual is $1.124 \times 10^{-10} M_\odot$.

All 114 archived pointwise control variants were re-executed using 58 distinct baseline densities. For Yukawa configurations, the exterior is shortened from 18L to 12L or extended to 24L, and the collocation tolerance is tightened from 2×10^{-7} to 5×10^{-8}. GR-IVP relative tolerances are tightened from 2×10^{-12} to 2×10^{-13}, also triggering tighter GR collocation. Own endpoint/peak configurations have separate tolerance tests.

Control	N	max $\lvert\delta M_{\mathrm{ADM}}\rvert$	max $\lvert\delta M_B\rvert$	max $\lvert\delta E\rvert$
Exterior extent 12L	24	4.608×10^{-12}	5.576×10^{-12}	1.729×10^{-9}
Exterior extent 24L	24	4.823×10^{-12}	6.439×10^{-12}	3.021×10^{-9}
Tighter Yukawa tolerance	24	1.520×10^{-10}	1.484×10^{-10}	5.223×10^{-7}
Tighter GR control	24	1.219×10^{-10}	2.173×10^{-10}	6.062×10^{-7}
Branch limit/peak: tighter tolerance	18	1.817×10^{-10}	6.055×10^{-10}	1.403×10^{-6}

Table 4: Pointwise numerical sensitivity from the supplied control tables; all 114 variants were re-executed. Mass columns are in solar masses and energy in 10^{51} erg. The baseline exterior extent is 18L. These are not physical uncertainty intervals or rigorous error bounds.

Domain changes alone shift ADM mass by at most $4.823 \times 10^{-12} M_\odot$; the largest shift including endpoint/tolerance

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controls is $1.817\times10^{-10}M_\odot$. Reproduced ADM changes differ from their stored values by at most $2.502\times10^{-12}M_\odot$. An additional quadrature of exterior scalar energy on the re-solved profiles agrees with $M_{\rm ADM}-m(R)$ to $6.526\times10^{-12}M_\odot$. These test finite-domain dependence at the selected points; they are not rigorous bounds over all densities and parameters.

The supplied 48/96-point baryon quadrature difference is at most $5.540\times10^{-12}M_\odot$. Independent first-law checks in native enthalpy coordinates give a largest relative finite-difference residual of 3.516×10^{-10}. The archived sequence-slope diagnostic $dM_{\rm ADM}/dM_B=e^{\nu_R-\phi_R}$ has residual at most 2.941×10^{-6}; it is not a mode calculation. A coarser/finer branch-only interpolation check gives maximum energy offsets from direct roots of 0.099695 and 0.013642×10^{51} erg, so tabulated contrasts use direct roots rather than these interpolants.

For each control perturbation, the empirical fixed-target diagnostic uses a linear projection back to the target:

$$\delta E_t=\delta E-\frac{dE/d\ell}{dM_t/d\ell}\,\delta M_t,\qquad \ell=\log_{10}\rho_{0c}. \tag{12}$$

Here M_t is the chosen matching mass and ℓ denotes the tabulated logarithmic density. The largest tested absolute projected changes of both pair members are summed, together with their quadrature shifts. Its maximum is $1.003\times10^{-6}\times10^{51}$ erg. This diagnostic is not a confidence interval. Fine local peak-fit outputs, with stored position changes up to 1.102×10^{-6} dex and mass changes up to $1.818\times10^{-10}M_\odot$, are included as archived-only checks: those fits were not re-executed in this assembly. The historical comparison against another source discretization has a baryon-mass discrepancy up to $1.083\times10^{-5}M_\odot$, outside the smaller internal collocation sensitivity; it is not assigned to a single error mechanism.

5 Interpretation and limitations

The useful result is a controlled distinction between energy ordering at fixed conserved baryon mass and baryon capacity at fixed asymptotic mass, for the stated local action and EOS convention. Positive fixed-M_B contrasts occur in the reported pairs and grow from the canonical to the selected near-limit states at $L=10$ km. This is not a claim of a new stellar phase or population.

The physical identification of the tabulated EOS with the bare $\varepsilon(n)$ in Eq. (1), the motivation and empirical validity of the coupling, and the interpretation of clocks/rods made of field-dependent matter remain open. Rotation, temperature, EOS uncertainty and radiation are not modelled or assigned uncertainty intervals. No coupled fluid–scalar perturbation problem or evolution was solved. Equal baryon mass between distinct theories does not specify initial/later states within one dynamical theory and cannot by itself predict emitted neutrino, scalar or gravitational-wave energy.

6 Data, reproducibility and disclosure

The accompanying package contains the unchanged numerical inputs and production code, bilingual editable TeX, figures, and `reproduce.py`. Primary values come from `data/comparisons.csv`; limits from `data/sequence_summary.csv`; numerical sensitivity from `data/convergence.csv` and the reproduced control tables. No remote input is required. Internal version YBE-2026-11-08-a is not a publication identifier. No DOI or repository record is assigned here, and identity of older archives is not inferred.

The source archive has SHA-256

`4ffe19e74db5f5c9237d244dcbc0e9a1a12b1e89dd1b13c27ec5684f8e3e47fe`.

Required-input hashes and their relative paths are in `metadata/provenance.json`. Re-execution reproduces the supplied algorithm; it is not independent physical validation or peer review.

Author-supplied disclosure: Numerical cross-checks with Academia Co-scientist (AI-assisted); not peer review.

This is the author's supplied disclosure, not a claim that a new Academia Co-scientist session was run while assembling this package.

Conclusion. Within the declared model, matched baryon mass reveals lower Yukawa ADM energy, while matched ADM mass reveals greater baryon capacity. Causality-limited matching and exterior-energy accounting are essential to interpreting this single equilibrium-energy result.

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